

SECTION 23 09 23

DIRECT DIGITAL CONTROL SYSTEM

TECHNICAL SPECIFICATION

Project	MERIDIAN-1 Data Centre, Hall A
Project code	MER-1-2026
Issue	Issued for Construction — Revision C
Date	17 February 2026

TABLE OF CONTENTS

PART 1 — GENERAL	
1.1	Summary
1.2	References
1.3	Definitions
1.4	Action Submittals
1.5	Informational Submittals
1.6	Closeout Submittals
1.7	Quality Assurance
1.8	Delivery, Storage, and Handling
1.9	Warranty
PART 2 — PRODUCTS	
2.1	Manufacturers
2.2	Performance Requirements
2.3	Equipment
2.4	Components
2.5	Controls and Interfaces
2.6	Source Quality Control
2.7	Factory Testing
PART 3 — EXECUTION	
3.1	Examination
3.2	Installation
3.3	Field Quality Control
3.4	Performance Verification
3.5	Commissioning
3.6	Training and Closeout

PART 1 — GENERAL

1.1 SUMMARY

- A. Provide a complete, coordinated system including engineering, manufacture, delivery, installation support, testing, commissioning, training, and closeout.
- B. Coordinate clearances, structural loads, electrical supplies, controls, fire strategy, access, and phased energisation.

1.2 REFERENCES

- A. Comply with the following referenced standards.
 - 1. ISO 16484-5:2022
 - 2. ASHRAE Guideline 36-2021
 - 3. IEC 61131-3:2013

1.3 DEFINITIONS

- A. “Approved” means accepted for the stated stage without relieving the Contractor of responsibility.
- B. “Provide” means design, supply, install, connect, test, commission, and document.

PART 1 — GENERAL

1.4 ACTION SUBMITTALS

- A. Submit project-specific product data, selected ratings, dimensions, weights, clearances, and connection requirements.
- B. Submit a clause-by-clause compliance statement identifying every exception.
- C. Submit certified calculations and performance data at scheduled conditions.

1.5 INFORMATIONAL SUBMITTALS

- A. Submit type-test certificates traceable to the offered construction.
- B. Submit quality certificates, calibration records, and proposed factory inspection and test plan.

1.6 CLOSEOUT SUBMITTALS

- A. Submit approved as-built data, final settings, test records, recommended spares, and operation and maintenance manuals.

PART 1 — GENERAL

1.7 QUALITY ASSURANCE

- A. Manufacturer shall maintain an ISO 9001-certified quality system and demonstrate comparable data-centre experience.
- B. Test instruments shall have current calibration traceable to a recognised laboratory.

1.8 DELIVERY, STORAGE, AND HANDLING

- A. Protect equipment from moisture, shock, corrosion, and contamination.
- B. Maintain equipment-tag, purchase-order, serial-number, and source-lot traceability.

1.9 WARRANTY

- A. Provide the project warranty commencing at Taking Over or the date stated in the Contract, whichever governs.

PART 2 — PRODUCTS

2.1 MANUFACTURERS

- A. Subject to compliance with the Contract Documents, provide the system by an approved manufacturer.
- B. Products shall be current production models with local technical support in India.

2.2 PERFORMANCE REQUIREMENTS

- A. Controller scan cycle shall not exceed **1 s**.
- B. Selections shall include stated tolerances, fouling allowances, harmonics, diversity, and redundancy as applicable.
- C. Published catalogue values shall not replace project-condition certified data.

PART 2 — PRODUCTS

2.3 EQUIPMENT

- A. Provide a complete factory-engineered assembly with coordinated ratings and interfaces.
 - 1. Native BACnet DDC controllers
 - 2. I/O modules, sensors and actuators
 - 3. Control panels, power supplies and engineering tools
- B. Materials shall be suitable for the operating environment and expected service life.
- C. Temperature sensor accuracy shall not exceed **0.15 C**.

2.4 COMPONENTS

- A. Components shall be accessible for inspection, isolation, removal, and replacement.
- B. Safety-critical devices shall be independently identified and fail to a safe state.
- C. Internal wiring, terminals, piping, and accessories shall be permanently identified.

PART 2 — PRODUCTS

2.5 CONTROLS AND INTERFACES

- A. Hardwired safety and shutdown functions shall not depend solely on the supervisory network.
- B. Trend retention shall be not less than **24 months**.
- C. Synchronise event timestamps to the project NTP service in Asia/Kolkata.

2.6 SOURCE QUALITY CONTROL

- A. Perform documented incoming inspection, in-process checks, and final inspection.
- B. Record material traceability, calibrated instrument numbers, torque values, and as-left settings.

PART 2 — PRODUCTS

2.7 FACTORY TESTING

- A. Submit the factory test procedure before testing.
- B. Test the offered rating and construction at the specified operating condition.
- C. Record raw measurements, calculated results, instrument details, and acceptance status.
- D. Do not claim equivalence for a different rating or condition without an approved engineering comparison.

PART 3 — EXECUTION

3.1 EXAMINATION

- A. Verify foundations, access routes, clearances, environmental conditions, and embedded services before installation.
- B. Do not proceed until unsatisfactory conditions are corrected.

3.2 INSTALLATION

- A. Install in accordance with approved drawings and manufacturer instructions.
- B. Install level, aligned, restrained, labelled, and accessible for operation and maintenance.
- C. Coordinate earthing, power, controls, penetrations, fire stopping, drainage, and builder's work.

PART 3 — EXECUTION

3.3 FIELD QUALITY CONTROL

- A. Inspect the completed installation and record deficiencies.
- B. Perform applicable continuity, insulation, pressure, leakage, rotation, interlock, and alarm tests.
- C. Close major deficiencies before functional performance testing.

3.4 PERFORMANCE VERIFICATION

- A. Demonstrate each specified performance criterion at stable scheduled conditions.
- B. Record measured values, instrument uncertainty, acceptance limits, and status.

PART 3 — EXECUTION

3.5 COMMISSIONING

- A. Point-to-point test coverage shall be **100 %**.
- B. Coordinate test prerequisites, witnesses, temporary loads, trend logs, and restoration.

3.6 TRAINING AND CLOSEOUT

- A. Train operations personnel using the installed equipment.
- B. Submit signed test records, final settings, training attendance, and approved closeout documents.

END OF SECTION